DecisionIQ — Dataset & Insights Reference Report

1. Project Context

DecisionIQ is an AI-powered Decision Intelligence Platform built around a synthetic multi-channel e-commerce business.

The dataset covers approximately January 2023 to December 2025 and combines transactional, customer, behavioral, return, and marketing data.

The goal of the dataset is to support business questions across four major areas:

- Sales and profitability
- Customer and acquisition analysis
- Website funnel analysis
- Marketing performance

DecisionIQ does not query only raw tables. A semantic SQL layer standardizes important business metrics before they are exposed to the AI system.

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2. Dataset Structure

The database contains eight primary tables.

customers

Approximate rows: 15,000

Important fields:

- customer_id
- customer_name
- age
- gender
- city
- state
- country
- signup_date
- acquisition_channel

The table represents customer-level information.

The `acquisition_channel` field identifies how a customer was originally acquired.

It should not be interpreted as the channel associated with every future purchase.

Typical acquisition channels include:

- Organic Search
- Google Ads
- Instagram
- Referral
- Email
- Direct

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products

Approximate rows: 500

Important fields:

- product_id
- product_name
- category
- subcategory
- cost_price
- selling_price
- supplier

The five major categories are:

- Electronics
- Home & Furniture
- Fashion
- Beauty
- Sports

The presence of both cost price and selling price allows DecisionIQ to calculate profitability rather than only revenue.

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orders

Approximate rows: 60,000

Important fields:

- order_id
- customer_id
- order_date
- order_status
- payment_method
- shipping_cost

Major statuses include:

- Delivered
- Returned
- Cancelled

An order marked as Returned does not necessarily mean that every product in the order was returned. At least one item in the order was returned.

Therefore, return analysis is primarily performed at the order-item level.

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order_items

Approximate rows: 179,000+

Important fields:

- order_item_id
- order_id
- product_id
- quantity
- unit_price
- discount_pct

This is the most important transaction-level table.

One order may contain several order-item rows.

Gross Sales is calculated as:

quantity × unit_price

Discounted Sales is calculated as:

quantity × unit_price × (1 − discount_pct)

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returns

Approximate rows: 16,400+

Important fields:

- return_id
- order_item_id
- return_date
- return_reason
- refund_amount

Each return corresponds to an individual order item.

Return-rate and refund analysis can therefore be performed at product and category level.

Important limitation:

Although a `return_reason` field exists, the frozen synthetic dataset does not provide sufficiently reliable reason-level patterns for strong causal claims.

Return rate and refund amount are reliable analytical variables.

Return reason should not be used to make statements such as:

“Fashion performs poorly because of sizing problems.”

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website_sessions

Approximate rows: 200,000

Important fields:

- session_id
- customer_id
- session_date
- device
- traffic_source
- pages_viewed
- session_duration_seconds
- added_to_cart
- converted

Customer ID may be NULL because anonymous visitors can browse the website.

Devices include:

- Desktop
- Mobile
- Tablet

This table allows DecisionIQ to analyze the funnel:

Session → Add to Cart → Conversion

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marketing_campaigns

Approximate rows: 80

Important fields:

- campaign_id
- campaign_name
- channel
- campaign_type
- start_date
- end_date
- campaign_budget

Marketing channels include:

- Email
- Google Ads
- Influencer
- Instagram
- Facebook
- YouTube

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campaign_performance

Approximate rows: 6,100+

Important fields:

- performance_id
- campaign_id
- date
- impressions
- clicks
- ad_spend
- conversions

- revenue_generated

This table supports campaign-performance metrics including CTR, conversion rate, CPA and ROAS.

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3. Database Relationships

The main transactional chain is:

customers
→ orders
→ order_items
→ products

Returns connect to:

order_items
→ returns

Website behavior connects through:

customers
→ website_sessions

Marketing performance connects through:

marketing_campaigns
→ campaign_performance

This means DecisionIQ can analyze the business from multiple perspectives while maintaining relational consistency.

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4. Core Business Metrics

Gross Sales

Gross Sales represents sales value before discounts and refunds.

Formula:

Gross Sales = quantity × unit_price

Dataset result:

₹6.605 billion

Discounted Sales

Discounted Sales represents revenue after discounts but before refunds.

Formula:

Discounted Sales = quantity × unit_price × (1 − discount_pct)

Dataset result:

₹5.900 billion

Refunds

Total customer refunds:

₹588.25 million

Net Revenue

Formula:

Net Revenue = Discounted Sales − Refunds

Dataset result:

₹5.312 billion

COGS

DecisionIQ uses adjusted COGS.

COGS is counted only for merchandise retained by customers.

If an item is returned, its COGS contribution is treated as zero in the analytical model because inventory is assumed to be recovered.

Dataset result:

₹4.242 billion

Gross Profit

Formula:

Gross Profit = Net Revenue − COGS

Dataset result:

₹1.069 billion

Gross Margin

Formula:

Gross Margin % = Gross Profit ÷ Net Revenue × 100

Dataset result:

20.13%

Average Order Value

DecisionIQ defines AOV as:

Discounted Sales ÷ non-cancelled orders

Dataset result:

approximately ₹109,354

This is a pre-refund AOV.

Website Conversion Rate

Formula:

Converted Sessions ÷ Total Sessions × 100

Dataset result:

24,986 conversions from 200,000 sessions.

Conversion Rate = 12.49%

Return Rate

DecisionIQ uses an item-level return rate.

Formula:

Returned Order Items ÷ Sold Order Items × 100

Dataset result:

approximately 10.18%

ROAS

Formula:

Attributed Marketing Revenue ÷ Ad Spend

CPA

Formula:

Ad Spend ÷ Campaign Conversions

DecisionIQ deliberately uses the term CPA rather than CAC.

A campaign conversion does not necessarily represent a newly acquired customer.

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5. Category Profitability

Electronics

Gross Profit: **₹236.79M**

Return Rate: **8.30%**

Gross Margin: **21.19%**

Electronics generates the highest total gross profit.

Home & Furniture

Gross Profit: **₹223.22M**

Return Rate: **9.97%**

Gross Margin: **19.62%**

Sports

Gross Profit: **₹221.19M**

Return Rate: **8.98%**

Gross Margin: **19.36%**

Beauty

Gross Profit: **₹218.44M**

Return Rate: **6.16%**

Gross Margin: **19.77%**

Beauty has the lowest return pressure among the five categories.

Fashion

Gross Profit: **₹169.49M**

Return Rate: **18.07%**

Refund Rate: **18.04%**

Gross Margin: **20.96%**

Fashion generates the lowest total gross profit.

However, Fashion does not have inherently poor gross margins.

Its gross margin percentage is actually relatively healthy.

The major distinguishing factor is its much higher return and refund pressure.

The correct interpretation is:

Fashion's weaker total profitability is associated with unusually high return and refund pressure rather than structurally poor retained-sale margins.

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6. Discount Analysis

Historical discount bands show a strong relationship between discount depth and profitability.

Approximate gross margins:

Discount	Gross Margin
0%	28.66%
5%	24.89%
10%	20.68%
15%	16.01%
20%	10.90%
25%	4.77%
30%	-2.04%

The major conclusion is:

30% discounts are loss-making in the historical dataset.

At 25%, profitability still remains positive but margin becomes extremely thin.

Return rate remains around 10% across most discount bands.

Therefore, the decline in margin is mainly associated with the economics of deeper discounting rather than a major increase in returns.

Important caveat:

The dataset does not establish that discounts caused changes in demand.

Discount levels should not be interpreted as demand drivers.

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7. Discount What-If Simulation

DecisionIQ also contains a scenario simulator.

This differs from the historical discount-band analysis.

The simulator applies one uniform discount level across the existing sales base while holding other factors constant.

Assumptions:

- Sales volume remains constant.
- Product mix remains constant.
- COGS remains constant.
- Refunds remain constant.
- Only the discount rate changes.

Approximate simulated margins:

Uniform Discount	Simulated Gross Margin
0%	29.49%
5%	25.39%
10%	20.79%
15%	15.59%
20%	9.65%
25%	2.81%
30%	-5.14%

At a 25% uniform discount:

Gross margin falls from approximately **29.49% to 2.81%** relative to the simulator's zero-discount baseline.

Gross profit falls by approximately **93%**.

At 30%, simulated gross profit becomes negative.

The simulator is a scenario model, not a forecast.

It does not account for possible demand increases caused by discounting.

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8. Website Funnel Insights

Total website sessions:

200,000

Total conversions:

24,986

Overall conversion:

12.49%

Device conversion rates:

Device	Conversion Rate
Desktop	15.16%
Tablet	12.00%
Mobile	11.26%

However, add-to-cart behavior is remarkably similar:

Device	Add-to-Cart Rate
Desktop	29.90%
Mobile	29.97%
Tablet	29.75%

Cart-to-conversion rates show a larger difference:

Device	Cart → Conversion
Desktop	50.70%
Tablet	40.33%
Mobile	37.57%

This creates one of the strongest DecisionIQ diagnostic findings.

Mobile users demonstrate approximately the same initial purchase intent as desktop users, but fewer complete purchases after reaching the cart.

Therefore:

The mobile conversion gap appears later in the funnel rather than during initial browsing or product-interest stages.

This does not prove that checkout UX is the cause.

Possible investigation areas include:

- checkout-step abandonment
- payment failures
- mobile performance
- form usability
- unexpected shipping or payment costs

These remain hypotheses until additional data is collected.

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9. Marketing Performance

Synthetic modeled marketing performance:

Channel	ROAS	CPA
Email	138.76x	₹23.05
Google Ads	47.31x	₹59.12
Influencer	32.33x	₹83.38
Instagram	17.41x	₹125.52
Facebook	11.73x	₹177.99
YouTube	7.10x	₹353.54

Within the synthetic scenario:

Email is the strongest modeled marketing channel.

YouTube is the weakest modeled channel.

Another important observation is spend allocation.

Email accounts for a relatively modest portion of total ad spend while producing a disproportionately large share of attributed revenue.

Instagram and YouTube consume significant spend but generate much smaller shares of attributed revenue.

The appropriate business recommendation is to review and test budget reallocation.

It would be incorrect to assume that moving all spend into Email would preserve the same ROAS because performance may not scale linearly.

Important limitation:

These are synthetic ROAS values.

They must not be presented as realistic industry benchmarks.

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10. Customer Acquisition Insights

Average revenue per acquired customer:

Acquisition Channel	Avg Revenue / Customer
Organic Search	₹360.6K
Instagram	₹359.2K
Direct	₹356.1K
Email	₹352.7K
Google Ads	₹347.5K
Referral	₹345.9K

The differences are relatively small.

Therefore, the strongest interpretation is not:

“Organic Search customers are dramatically more valuable.”

Instead:

Customer quality is broadly similar across acquisition channels.

Organic Search's higher total contribution is largely influenced by having more acquired customers.

This highlights the difference between:

- total contribution
- per-customer economics

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11. Geographic Analysis

State-level gross margins are generally around:

20%

Return rates are broadly around:

10%

Maharashtra generates the highest overall profit largely because it contains more customers and transactions.

There is no strong evidence that customers in Maharashtra have fundamentally superior unit economics.

The dataset does not reveal a major geographic profitability anomaly.

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12. Monthly Trend

Revenue and profit increase strongly from 2023 through 2025.

However, this trend must be interpreted carefully.

Customers sign up progressively during the dataset period.

Customers cannot place orders before their signup date.

As a result, later months naturally contain a larger potential customer base.

This creates a synthetic upward boundary effect.

Therefore, it would be misleading to claim:

“The company experienced explosive organic growth.”

A safer interpretation is:

The dataset contains a strong upward revenue trend, but part of that trend is structurally generated by customer signup timing.

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13. Anomaly Detection

DecisionIQ detects anomalies based primarily on unusual month-over-month movements rather than raw revenue levels.

This avoids flagging every later month merely because the synthetic dataset has an upward trend.

Three notable anomalies were identified.

May 2024

Refund growth:

+51.65% month over month

Refunds were flagged as statistically unusual.

Revenue and gross profit also increased, but did not cross the anomaly threshold.

February 2025

Net revenue:

−5.95% month over month

Net revenue was flagged as an unusual negative movement.

Gross profit also declined but did not cross the anomaly threshold.

December 2025

Net revenue:

+48.28% MoM

Gross profit:

+49.59% MoM

Refunds:

+40.58% MoM

All three were strongly flagged.

However, December 2025 must be interpreted cautiously because the dataset's synthetic boundary effect contributes to unusually high late-period activity.

Statistical anomaly does not imply a real-world causal event.

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14. DecisionIQ Semantic Layer

DecisionIQ primarily exposes four business-ready views to the AI.

vw_order_item_metrics

Used for transactional and profitability analysis.

Important fields include:

- category
- subcategory
- gross_sales
- discounted_sales
- refund_amount
- net_revenue
- cogs
- gross_profit
- is_returned

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vw_customer_metrics

Used for customer-level analysis.

Important fields include:

- acquisition_channel
- total_orders
- net_revenue
- gross_profit
- total_items
- returned_items
- return_rate_pct
- avg_order_value

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vw_marketing_metrics

Used for marketing analysis.

Important fields include:

- channel
- campaigns
- impressions
- clicks
- conversions
- ad_spend
- attributed_revenue
- ctr_pct
- click_conversion_rate_pct
- cpa
- roas

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vw_website_funnel

Grain:

device × traffic_source

Important fields include:

- total_sessions
- add_to_cart_sessions
- converted_sessions
- add_to_cart_rate_pct
- conversion_rate_pct

- cart_to_conversion_pct
- avg_pages_viewed
- avg_session_duration_seconds

The semantic layer ensures that KPI formulas are consistent across AI-generated queries.

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15. DecisionIQ Analytical Capabilities

The current system supports several types of analytics.

Descriptive Analysis

Example:

“Which marketing channel has the highest ROAS?”

The AI generates SQL and queries the semantic layer.

Diagnostic Analysis

Example:

“Why is mobile conversion lower than desktop?”

DecisionIQ routes the question to deeper Python-based funnel analysis.

Profitability Driver Analysis

Example:

“Why is Fashion less profitable?”

DecisionIQ analyzes gross profit, gross margin, refunds and return rates.

Scenario Analysis

Example:

“What happens if discounts increase to 25%?”

DecisionIQ invokes the discount what-if simulator.

Anomaly Detection

Example:

“Were there any unusual revenue or profit movements?”

DecisionIQ analyzes unusual month-over-month changes.

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16. Major Dataset Limitations

Three limitations should always be remembered.

Synthetic data

The data is analytically designed rather than generated from a real company.

Patterns should not be presented as industry benchmarks.

Return reasons

Return rates and refund values are useful.

Return-reason distributions are not sufficiently reliable for causal conclusions.

Time-series boundary effect

Customers enter the dataset gradually.

Later periods have more customers capable of placing orders.

Therefore, some apparent growth is structurally induced.

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17. Six Insights to Memorize

For interviews or demos, the following six findings are the most important:

1. Electronics generated the highest gross profit at approximately ₹236.8M.

2. Fashion generated the lowest total profit and had an approximately 18% return rate, far above the other categories.

3. Historical 30% discounts were loss-making, while 25% discounts left only very thin margins.

4. Desktop converted at approximately 15.2% compared with Mobile at 11.3%, despite nearly identical 30% add-to-cart rates.

5. Email produced the highest modeled ROAS, while YouTube produced the lowest within the synthetic marketing scenario.

6. Customer economics were relatively similar across acquisition channels, meaning volume often explained total contribution differences better than per-customer value.

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18. 60-Second Dataset Explanation

DecisionIQ uses a synthetic three-year e-commerce dataset covering customers, products, orders, order items, returns, website sessions, marketing campaigns and campaign performance.

The transactional layer contains roughly 60,000 orders and around 180,000 order items, while the behavioral layer includes 200,000 website sessions and the marketing layer includes approximately 80 campaigns.

I created a SQL semantic layer on top of the raw data to standardize metrics such as net revenue, adjusted COGS, gross profit, return rate, website conversion, CPA and ROAS.

This allows the AI to query consistent business-ready metrics instead of recreating formulas for every question.

The dataset supports descriptive, diagnostic, scenario and anomaly analysis. Key findings include strong return pressure in Fashion, significant margin erosion at deep discounts, lower mobile post-cart conversion, and large differences in modeled marketing efficiency.

Because the dataset is synthetic, I explicitly account for limitations such as unrealistic marketing ROAS values, unreliable return-reason causality and a time-series boundary effect caused by customers signing up progressively over the dataset period.